Quechua → Spanish Machine Translation

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Abstract

The goal of this project is to create a machine translation system for Quechua → Spanish translation. To train a machine translation model, a parallel corpus is required—in which a text appears in two languages. Given that Quechua is a low-resource language, there were not a lot of datasets readily available. We were able to find a Quechua-Spanish parallel corpus on Hugging Face that contains translations from Spanish to Ayacucho Quechua. Finding this corpus was one of the main reasons that we decided to have our translation model be Quechua to Spanish instead of to English or any other languages. Also, Spanish is a second language for many native Quechua speakers, and the most spoken language of the countries that Quechua is native to. Quechua-Spanish machine translation could help to preserve indigenous knowledge, make services more accessible, and bridge communication gaps.

1 Introduction

Quechua is a pre-Incan language having originated approximately 4,500 years ago. It's believed at one point Quechua was spoken in "Perú, Bolivia, Ecuador [and parts of] Colombia" (Huaman, 2024**)**. It was integral in the Incan Empire as it united lands from the coast to the Andes ("Descubre el origen del quechua, el idioma de los incas", 2021).

The arrival of the Spanish in 1526 in Northern Peru marked the beginning of colonization and subjugation of the Incan people at the hands of Spanish conquistador Franscisco Pizzaro ("The Inca Empire, Spanish Conquest, and Independence"). The culture and freedom of the indigenous people became under threat, including the language of Quechua. A language with such rich history and evolution became an endangered language due to the atrocity of colonization by the Spanish.

Therefore, in order to aid a higher level of accessibility to learning the language and its preservation, we sought to create a natural language model that can facilitate those objectives. We used both a baseline traditional approach and a large language model, to create systems that could perform machine translation from Quechua to Spanish.

For our specific model we fine-tuned a pre-trained T5-small model from the Hugging Face Transformers Library. We also conducted few-shot prompting with ChatGPT and asking Gemini to do the translations. Our fine-tuned model was outperformed by that of the LLM's machine translation abilities.

2 Data

2.1 Overview

The dataset selected to complete this project for data training is available on the Hugging Face platform. It has side by side Spanish phrases and the corresponding Quechua phrases. There is 103k rows in the dataset, and it was collected from an array of different online sources. These sources include educational apps made to teach Quechua, stories translated from Spanish to Quechua, and more. The specific dialect of this language is Ayacucho Quechua. The data that we used in order to perform our evaluations was retrieved from the Pontificia Universidad Católica Del Perú. The textbook is titled, Qayna, Kunan, Paqarin by Roberto Zariquiey and Gavina Córdova.

2.2 Limitations

One limitation of the data was that the dataset is relatively small in comparison to high-resource languages like English-German, English-Mandarin, etc. This limitation did not affect our project as we ended up using a subset of the data, but for a truly robust model to be trained, more data is usually better and can improve a model's ability to generalize unseen or more complex phrases.

Another limitation is that the dataset is specifically Ayacucho Quechua, which is a dialect of Quechua spoken in the Ayacucho region of Peru. This means that the model trained on the data may not be representative of all dialects of Quechua equally.

3 Methods

3.1 Baseline Model Training

For the task of machine translation we fine tuned a pre-trained T5-small model from the Hugging Face Transformers Library. T5 is an encoder-decoder transformer model that treats NLP tasks as text-to-text problems, converting every NLP task into a text generation task, and making it a good choice for translation tasks.

To make training faster and manageable with our resources, we limited the dataset to 10,000 examples. This subset was enough to fine-tune the model and get reasonable translation output while avoiding excessive computational cost and long training time.

The next step was preprocessing the data. The Quechua input was prefixed with "translate Quechua to Spanish: " because the T5 model requires a prompt telling it what task to preform. After training the model, it also requires the same kind of input, with the prefixed prompt.

After loading the T5-small model and its corresponding tokenizer, the Quechua input phrases and corresponding Spanish target phrases were tokenized with a maximum length of 28 tokens before they are truncated.

After preprocessing by prefixing, tokenizing, and truncating, parameters for training were set and managed using Hugging Face's Seq2SeqTrainer. We kept all but one of the default parameters, changing the number of epochs to 5, meaning that the model saw each training example five times. We chose to increase the number of epochs because it ensures that the model revisits the data multiple times to learn better. We also used an 80/20 split for training and evaluation sets. Given that we had 10,000 examples the size of the training and evaluation sets were 8,000 and 2,000 respectively.

Versions of the model are saved intermediately at checkpoints, after a certain amount of training steps (500, 1000, and 2500). We selected checkpoint-2500 for testing and evaluating since it was the last checkpoint, and thus expected it to represent the best-trained version of the model.

After training the model, we tested it by performing the translation task, prompting the model with a set of Quechua phrases with the prompt "translate Quechua to Spanish: " and looping through those prompts to get the translated output. This output was then compared to reference Spanish translations in our evaluation.

3.2 BLEU Evaluation

For our first evaluation method, we employed the usage of the BLEU (Bilingual Evaluation Understudy) metric. This is algorithm that is used to evaluate the quality of text that has been translated from one language to another using machines. Quality is measured in how close the machine translated text is to how a "professional human" would translate it ("Metric: bleu"). Scores are found through comparing parts of the text produced by the model to high quality translations. It does this specifically by "comparing the n-grams" of the reference translations to the model's generated output ("NLP – BLEU Score for Evaluating Neural Machine Translation – Python", 2024). We conducted this evaluation for our baseline model, the responses from Few-Shot prompting with ChatGPT, and Gemini.

Due to the popularity of this metric to evaluate the quality machine-translated text, there is multiple online libraries that can be used to perform the comparison. For this specific project, we chose to use the method available by the Natural Language Toolkit ("[NLTK] Documentation"). All that was necessary to perform is to process the machine created texts and the high-quality reference material into segments (Verma, 2022). After this, we iterated through both the model's output and the reference translation and used NLTK's offered function to calculate the BLEU score (Verma, 2022). We performed this step for

ten different phrases calculating the score for each comparison.

3.3 Evaluation 2

Our second evaluation method was chrF (CHaRacter-level F-score). This metric looks at character sequences by calculating the F-score using the average of n-gram precision and recall. There is also a weight component that favors recall more than precision, and its default is 2.

$$chrF = (1 + \beta^2)\frac{ngrP \cdot ngrR}{\beta^2 \cdot ngrP + ngrR}$$

This essentially means that precision is the matching charact n-grams divided by the total character n-grams in the hypothesis. Recall is the matching character n-grams divided by the total character n-grams in reference. Beta controls how important recall is in comparison t precision. The chrF score ranges from 0 to 100, the higher the score, the closer the hypothesis is to the reference.

4 Results

4.1 Baseline Model Training

The training loss after each epoch was as follows:

Epoch	Training Loss
1	2.93
2	2.91
3	2.89
4	2.88
5	2.87

The training loss did decrease gradually, although by minute amount, after each epoch (from 2.93 at epoch 1to 2.87 at epoch 5).

The evaluation loss was also calculated after each epoch and was as follows:

Epoch	Evaluation Loss
1	2.65
2	2.63
3	2.62
4	2.618
5	2.616

The evaluation loss, similar to the training loss, also improved with later epochs, from 2.65 at epoch 1 to 2.61 at epoch 5.

4.2 BLEU Evaluation

The BLEU score that was calculated was done so for our baseline model, the few shot prompting with ChatGPT, and Gemini. The BLEU score for all three were incredibly low, with them all being close to 0 for the averages ("[NLTK] Documentation"). Specifically, for our baseline model the average BLEU score was 0.0, for the few-shot prompting with ChatGPT it was 0.0, and for Gemini it was 3.1204474728177387e-232. This is indicative that the model's abilities to the translation were subpar and unable to compete with what a professional human would have translated.

4.3 chrF Evaluation

The chrF score was calculated for our model's translations, chatGPT's translations, and Gemini's translations. Compared to ChatGPT and Gemini, our model received a much worse chrF score. Gemini outperformed all three models, with ChatGPT coming in behind, and our model falling in last place.

	ChatGPT	Gemini	Our model
Phrase 1	42.27	85.64	8.18
Phrase 2	13.30	76.96	10.96
Phrase 3	15.73	38.55	8.02
Phrase 4	17.50	62.85	13.19
Phrase 5	12.19	54.34	9.30
Phrase 6	32.86	55.42	29.32
Phrase 7	29.96	51.77	20.67
Phrase 8	39.94	49.05	16.80
Phrase 9	10.44	67.02	6.49
Phrase 10	8.76	85.22	12.04

5 Discussion

5.1 Baseline Model Training

The training loss and evaluation loss decreased across epochs, showing improvement in learning, albeit slow. The loss results indicate moderate but consistent learning across epochs and shows that small improvements were made with each epoch.

Overall, the results of the efficacy of the model, through the decrease in training and evaluation loss results show that the model is learning patterns in the data gradually.

5.2 BLEU Evaluation

Given the values produced by the BLEU scoring metric, all three methods appeared subpar in comparison to professional human translation. The closer to 0 that translates to their being no overlap "for any order of n-grams" ("[NLTK] Documentation"). It would appear that the Gemini Translate BLEU score was the highest, but the value it produced is so minimal that it is a negligible difference. Henceforth, using the BLEU metric, our baseline model, the LLM model, and Gemini did not do a comparable job at all to what a human translator would have produced.

5.3 chrF Evaluation

As the table shows, Gemini's chrF score was higher for each phrase compared to our model and ChatGPT's. Therefore, according to chrF, Gemini has the highest quality translations from Quechua to Spanish.

6 Future Work

In the future, we would like to explore alternative approaches to machine translation such as using larger pre-trained models or using more advanced and robust fine-tuning. We would also have liked to train with more data and look at other various forms of Quechua dialects.

7 Limitations

Due to the wide range of places that the language is spoken, there are difference existing dialects. As a result of this, we narrowed to focus on one specific dialect, that of Ayacucho or Chanco. However, there is not a lot of resources online for Quechua, a lot of it is just information for tourists to find information to "connect with locals". Additional to this, a lot of resources do not state what dialect or region the Quechua is from. This is a great problem as it would make our models have conflicting information for what means what in Quechua. Henceforth, the number of available resources for Quechua is already slim, and that of a specific dialect is even less. This limited the amount of access to data we had, and especially for data that we could use to evaluate the accuracy of our models. We ended up finding a textbook used for the specific dialect, but it took some time and care in order to find it.

Another limitation is the amount of computation power and time needed to process the amount of data we used for our baseline model. We ended up being able to train on 10,000 data points, but this took time to figure out and make possible. This sort of training requires higher resources than we available to us via our computers, so we had to perform it on the CRC's network. This was a limitation in the allotted amount of time, which held us back from being able to train the model on more points while we acquainted ourselves with the process.

8 Ethical Issues

One ethical issue may be how the Quechua data is collected. If it is collected without consent than there could be the issue of __. Additionally, the data collected could disrespect indigenous ownership of language and knowledge.

Mistranslation / misrepresentation of Quechua could also occur with automated translation. This could lead to erasure of sacred concepts and simplifying things in the Quechua language. This could potentially lead to stereotypes being perpetuated and other cultural narratives being degraded. Additionally, biases towards colonial / urban Spanish might be reinforced, deepening the inequalities of how Quechua speakers are represented.

Another ethical issue is with Quechua being an oral language, text translations might lead to oral knowledge being neglected and/or erased.

9 Group Member Task Breakdown

Vibha handled the task of training and fine-tuning the baseline machine translation model. She filled out multiple parts of the report including the baseline model sections, abstract, data limitations, and future work.

Raquel handled the tasks of the BLEU evaluation for the baseline model, the ChatGPT few shot prompting, and the Gemini responses. She additionally conducted the few shot prompting for ChatGPT for translating quechua phrases to Quechua. She additionally filled out multiple parts of the report including the

introduction, the evaluation parts of the report, and the limitations.

Vaageesha handled the task of the chrF evaluation for the baseline model, ChatGPT, and Gemini responses. She additionally filled out the chrF evaluation and future work sections of the report.

All of our code used to train and evaluate the models can be accessed at the following GitHub link: https://github.com/vibha-h/quechua-spanish-mt

Acknowledgments

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A Appendices

Appendices are added after the References section by restarting the header numbering using style "A, B, C".

B Supplementary Material

Supplementary material also be included with the Appendices.